Stratified Randomized Experiments

Zhichao Jiang

Sun Yat-sen University

Spring 2026

Setup

- SRE: CRE within each stratum $X_i = k$

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n_{[k]} &= \#\{i : X_i = k\}, \quad \pi_{[k]} = n_{[k]}/n \\
n_{[k]1} &= \#\{i : X_i = k, Z_i = 1\}, \quad n_{[k]0} = \#\{i : X_i = k, Z_i = 0\}
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- Compute the p-value based on the randomized distribution of treatment assignment
  - Exact distribution in small samples
  - Large-sample approximation
  - Monte Carlo approximation as a general strategy

Choices of test statistics

- Stratified estimator

$$
\begin{aligned}
\hat{\tau}_S &= \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]} \\
\hat{\tau}_{[k]} &= n_{[k]1}^{-1} \sum_{i=1}^{n} \mathbf{1}(X_i = x, Z_i = 1) Y_i - n_{[k]0}^{-1} \sum_{i=1}^{n} \mathbf{1}(X_i = x, Z_i = 0) Y_i
\end{aligned}
$$

- Studentized stratified estimator

$$
\begin{aligned}
t_S &= \frac{\hat{\tau}_S}{\sqrt{\hat{V}_S}} \\
\hat{V}_S &= \sum_{k=1}^{K} \pi_{[k]}^2 \left(\frac{\hat{S}_{[k]}^2(1)}{n_{[k]1}} + \frac{\hat{S}_{[k]}^2(0)}{n_{[k]0}} \right)
\end{aligned}
$$

Neymanian inference

- SRE is a CRE within each stratum

- Stratum-specific difference in means $\hat{\tau}_{[k]}$ is unbiased for $\tau_{[k]}$ with variance

$$\text{var}(\hat{\tau}_{[k]}) = \frac{S^2_{[k]}(1)}{n_{[k]1}} + \frac{S^2_{[k]}(0)}{n_{[k]0}} - \frac{S^2_{[k]}(\tau)}{n_{[k]}}$$

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- Stratified estimator $\hat{\tau}_S = \sum_{k=1}^{K} \pi_{[k]} \hat{\tau}_{[k]}$ s unbiased for τ with variance

$$\text{var}(\hat{\tau}) = \sum_{k=1}^{K} \pi^2_{[k]} \text{var}(\hat{\tau}_{[k]})$$

Neymanian inference

- Conservative variance estimator

$$\hat{V}_S \quad = \quad \sum_{k=1}^{K} \pi_{[k]}^2 \left(\frac{\hat{S}_{[k]}^2(1)}{n_{[k]1}} + \frac{\hat{S}_{[k]}^2(0)}{n_{[k]0}} \right)$$

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- CLT for $\hat{\tau}_S$: Liu and Yang (2020)
- Conservative $1-\alpha$ confidence interval

$$\hat{\tau}_S \pm z_{1-\alpha/2}\sqrt{\hat{V}_S}$$

Comparing SRE with CRE

- Estimators: $\hat{\tau}$ in CRE, $\hat{\tau}_S$ in SRE

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- Estimators: $\hat{\tau}$ in CRE, $\hat{\tau}_S$ in SRE

- Setting for comparison: $e_{[k]} = e$ for all k

- $\hat{\tau} = \hat{\tau}_S$
- Compare $\text{var}_{\text{CRE}}(\hat{\tau})$ with $\text{var}_{\text{SRE}}(\hat{\tau}_S)$

Variance comparison

- Variance terms under CRE

$$
\begin{aligned}
S^2(1) &= (n-1)^{-1}\sum_{i=1}^{n}\{Y_i(1)-\bar{Y}(1)\} \\
&= \sum_{k=1}^{K}\left[\frac{n_{[k]}-1}{n-1}S^2_{[k]}(1)+\frac{n_{[k]}}{n-1}\left\{\bar{Y}_{[k]}(1)-\bar{Y}(1)\right\}^2\right] \\
S^2(0) &= \sum_{k=1}^{K}\left[\frac{n_{[k]}-1}{n-1}S^2_{[k]}(0)+\frac{n_{[k]}}{n-1}\left\{\bar{Y}_{[k]}(0)-\bar{Y}(0)\right\}^2\right] \\
S^2(\tau) &= \sum_{k=1}^{K}\left\{\frac{n_{[k]}-1}{n-1}S^2_{[k]}(\tau)+\frac{n_{[k]}}{n-1}(\tau_{[k]}-\tau)^2\right\}
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- With large $n_{[k]}$'s

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\begin{aligned}
\mathrm{var}_{\mathrm{CRE}}(\hat{\tau}) &= \frac{S^2(1)}{n_1}+\frac{S^2(0)}{n_0}-\frac{S^2(\tau)}{n} \\
&\approx \sum_{k=1}^{K}\left\{\frac{\pi_{[k]}}{n_1}S^2_{[k]}(1)+\frac{\pi_{[k]}}{n_0}S^2_{[k]}(0)-\frac{\pi_{[k]}}{n}S^2_{[k]}(\tau)\right\} \\
&\quad+\sum_{k=1}^{K}\left[\frac{\pi_{[k]}}{n_1}\left\{\bar{Y}_{[k]}(1)-\bar{Y}(1)\right\}^2+\frac{\pi_{[k]}}{n_0}\left\{\bar{Y}_{[k]}(0)-\bar{Y}(0)\right\}^2-\frac{\pi_{[k]}}{n}(\tau_{[k]}-\tau)^2\right]
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- With large $n_{[k]}$'s, the difference between $\mathrm{var}_{\mathrm{CRE}}(\hat{\tau})$ and $\mathrm{var}_{\mathrm{SRE}}(\hat{\tau}_S)$

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\sum_{k=1}^{K} \pi_{[k]} \left[\frac{\left\{ \bar{Y}_{[k]}(1) - \bar{Y}(1) \right\}^2}{n_1} + \frac{\left\{ \bar{Y}_{[k]}(0) - \bar{Y}(0) \right\}^2}{n_0} - \frac{(\tau_{[k]} - \tau)^2}{n} \right] \geq 0
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  - equality holds only if $\frac{\sqrt{n_0}}{\sqrt{n_1}}\{\bar{Y}_{[k]}(1) - \bar{Y}(1)\} + \frac{\sqrt{n_1}}{\sqrt{n_0}}\{\bar{Y}_{[k]}(0) - \bar{Y}(0)\} = 0$

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  - SRE is more efficient than CRE if X is predictive of the potential outcomes
  - Block what you can and randomize what you cannot — Box *et al.*

Post-stratification in a CRE

- Can we analyze a CRE with a discrete X by treating it as a SRE
  - $\boldsymbol{n} = \{n_{[k]1}, n_{[k]0}; k = 1, \ldots, K\}$ is fixed in SRE but random in CRE
  - Conditional inference given $\boldsymbol{n}$

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- Duals for using X
  - design stage: SRE
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  - design stage: SRE
  - analysis stage: post-stratification
  - small difference asymptotically

Remarks

- How do we choose X when design an SRE?
  - X should be predictive of the outcome
  - based on logistical convenience
- How do we choose K?
  - more stratification increases efficiency with large enough strata
  - extremly large K may decrease efficiency
- What if X is continuous?
  - discretization
  - rerandomization and covariate adjustment